Few-Shot Object Detection With Multilevel Information Interaction for Optical Remote Sensing Images

Lefan Wang, *Graduate Student Member, IEEE*, Shaohui Mei, *Senior Member, IEEE*, Yi Wang, *Member, IEEE*, Jiawei Lian, *Graduate Student Member, IEEE*, Zonghao Han, and Xiaoning Chen

***Abstract*— Metalearning has been widely applied to solve the few-shot object detection (FSOD) problem in natural scenes, which performs similarity measurement and information aggregation of the support set and the query set. However, regarding remote sensing images (RSIs), many difficulties caused by their disparities need to be further addressed, such as inconsistencies in imaging scale, direction, and background between support and query images. These result in feature misalignment and attention bias, interfering with model performance. In this article, a multilevel information interaction (MLII) strategy is proposed for FSOD to alleviate feature misalignment and attention bias. Information interactions are conducted within multiple scales of features and highlight similar regions of query and support features. A semantic enhancement module (SEM) is proposed to assist MLII in extracting key information and achieving more discriminative feature representation. Moreover, a feature cross-aggregation module (FCM) with separate classification losses is designed to train the detector to identify objects that coexist in query and support images. Extensive experiments demonstrate that the proposed method outperforms several state-of-the-art few-shot object detectors over commonly used benchmark datasets, i.e., DIOR and NWPU-10.**

***Index Terms*— Few-shot learning (FSL), metalearning, object detection, optical remote sensing images (RSIs).**

I. Introduction

REMOTE sensing images (RSIs) combined with deep learning have emerged as a crucial tool for Earth spatial information exploration, particularly in the domain of optical remote sensing object detection [1], [2], [3], [4], [5], [6]. However, deep learning-based methods are data-driven and require training on extensive datasets. Generally, obtaining large-scale RSI datasets is extremely difficult, especially for RSIs over broad ground areas [7]. In many applications such as object detection, semantic segmentation, and so on, some particular targets of interest are so rare that they are hard to capture by imaging devices [8], [9], [10], [11], [12]. Besides, data annotation is extremely labor-intensive and time-consuming. Consequently, the lack of annotated data leads to overfitting, poor model performance, and generalizability, making it challenging to train effective object detection models. Few-shot object detection (FSOD) is proposed to solve these problems on RSIs, which efficiently extends the learned general knowledge from a data-rich base dataset to novel classes with limited labeled data [13], [14], [15], [16].

Few-shot learning (FSL) in remote sensing faces unique challenges due to the limited availability of labeled data and the complexity of Earth observation tasks. Remote sensing datasets often contain high-dimensional and large-scale imagery, making traditional FSL approaches less effective. In addition, remote sensing tasks involve diverse environmental conditions, seasonal variations, and sensor characteristics, further complicating the learning process. To address these challenges, researchers have explored various techniques, including transfer learning [17], [18], [19], metric learning [20], [21], and metalearning [21], [22]. Transfer learning [23], [24], [25] leverages pretrained models on large-scale datasets to extract generic features, which can be fine-tuned on smaller remote sensing datasets to adapt to specific tasks. Metric learning [26], [27] assumes that a model can measure the similarity of two pictures in a certain domain, and then, given a sample of the target domain, the most similar instance can be found. The metric learning relies on the hypothesis that object features in a query image should have high similarity with prototypes of the corresponding category. However, due to the low-sample scenario, the query set and the support set may not have high similarity. Metalearning approaches [28], [29] aim to learn from multiple few-shot tasks to improve the model's generalization ability, enabling it to adapt quickly to new tasks with minimal labeled data.

FSOD needs to recognize novel objects using a few annotated examples (i.e., support set) and localize objects in the query image. Existing works can be mainly grouped into the following two categories according to the training paradigm.

1) Single-branch methods [30], [31], [32] grounded in the principles of transfer learning are founded upon a typical object detection algorithm, such as Faster R-CNN [33], YOLO [34], [35], and so on [36], [37]. To enhance

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Lefan Wang, Shaohui Mei, Jiawei Lian, Zonghao Han, and Xiaoning Chen are with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an 710129, China (e-mail: meish@nwpu.edu.cn).

Yi Wang is with the Department of Electrical and Electronic Engineering, Hong Kong Polytechnic University, Hong Kong (e-mail: yi-eie.wang@polyu.edu.hk).

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the performance of knowledge transfer, other strategies are employed, such as knowledge distillation [38], knowledge inheritance [39], contrastive learning [40], and decoupling classifier heads [41]. These single-branch methods tend to prioritize learning discriminative representations of objects and are prone to overfitting under few-shot scenarios.

2) Two-branch methods [42], [43], [44], [45], [46], [47], [48] apply metric learning and metalearning ideas to FSOD and use episodic training through a Siamese network to acquire task-level knowledge in parallel on base classes and generalize better to novel classes. Previous works have demonstrated the advantages of the two-branch methods, such as feature fusion [43], [44], [49], feature alignment [42], GCN [50], and nonlocal attention/transformer [51], [52], [53], [54]. Consequently, we shift our attention to the two-branch methods.

Due to the predominant composition observed in natural scene images, characterized by objects predominantly positioned at the center of the frame with minimal variations in size and location, the discernment of foreground objects is notably facilitated. However, RSIs contain objects characterized by random orientations, dense arrangements, and varying scales. Existing two-branch methods of FSOD on RSIs endeavor to address the unique challenges posed by RSIs by introducing multiscale framework [29], prototype-guided region proposal network (P-G RPN) [55], and so on. Metalearning methods for FSOD on RSIs [29] develop a multiscale object detection framework upon the YOLOv3 architecture. Prototype-CNN [55] used class-aware P-G RPN and extended the head of faster region-based CNN (R-CNN). The previous two-branch methods on RSIs have made structural improvements, and they primarily interact at the high-level features (e.g., alignment and soft weighting) for similarity learning. However, they neglect the feature misalignment and attention bias between the support set and the query set.

In the FSOD methods for natural scene images, the support set is generally formed through two distinct approaches: 1) cropping with padding surrounded pixels (e.g., 8 px and 16 px) and then resizing [42], [47], [52], [56] and 2) adding a mask [43], [44]. Following this paradigm, the remote sensing support images are constructed identically [29], [55]. However, this brings obvious problems such as: 1) significant appearance inconsistency; 2) feature misalignment; and 3) attention bias. As shown in Fig. 1, significant inconsistencies regarding object pose, scale, lighting, orientation, and background exist, even when the query and support images are constructed from the same source image. The substantial disparity in scale between query and support images may lead to feature misalignment after feature extraction by shared-parameter Siamese networks, posing challenges to exploring the similarity relationships fully [57]. The complexity of the query image (multitarget, multicategory, complex background) is much higher than that of the single-object support image, resulting in unequal attention to the query and support features. In the case of an "airplane," the query image's attention falls on all the airplane regions, and the cropping support image is on the airplane's easily recognizable head and tail. Furthermore, when there are multiple targets in the query image, due to the few training novel class samples, attention tends to focus more on the base classes than the novel classes.

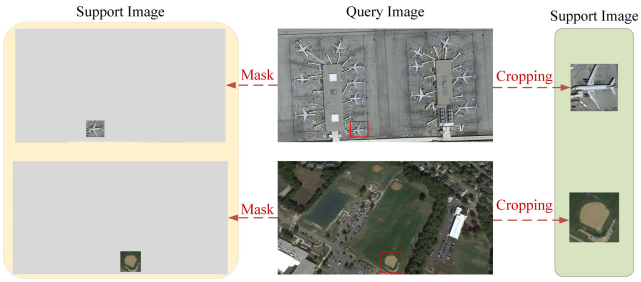


Fig. 1. Display of two main ways of support set construction and inconsistencies between support and query images.

In this article, a novel multilevel information interaction (MLII) is proposed in the FSOD framework for optical RSIs to alleviate the abovementioned problems. The proposed MLII model aims to align features of different scales from the two branches and adjust the attention to emphasize mutual elements more strongly. Specifically, the MLII consists of two shallow interactions and one deep interaction. Shallow interactions adjust the attention bias with similarity measurements on key maps of two branches to decide which portion of the features to align. Value maps that sufficiently preserve distinct support features are aligned and fused into the query value maps to update the query branch. Thus, shallow interactions serve to align features and adjust attention biases. The deep interaction, occurring before the region proposal network (RPN), not only aligns key information but also adaptively learns the correspondence between region similarities of the two branches. The semantic enhancement module (SEM) enhances important channels to assist the MLII in extracting key information, contributing to a more comprehensive and discriminative feature representation. Furthermore, to facilitate the detector's recognition of coexisting objects in query and support images, support features are divided into positive and negative support. Subsequently, the feature cross-aggregation module (FCM) individually fuses the positive and negative class support features with RoI features. Introducing separate classification losses allows the training of fusion features to address the issue of ambiguity in distinguishing different categories.

In summary, the main contributions of this article are given as follows.

1) An MLII-based FSOD is proposed for optical RSIs to eliminate feature misalignment and attention bias of query set and support set caused by significant appearance inconsistency. This effectively improves the FSOD performance.
2) An SEM is introduced to strengthen the key feature extraction ability of the MLII and achieve a more discriminative feature representation.
3) An FCM is devised to separately fuse the positive and negative class support features with RoI features, followed by the proposed separate classification losses

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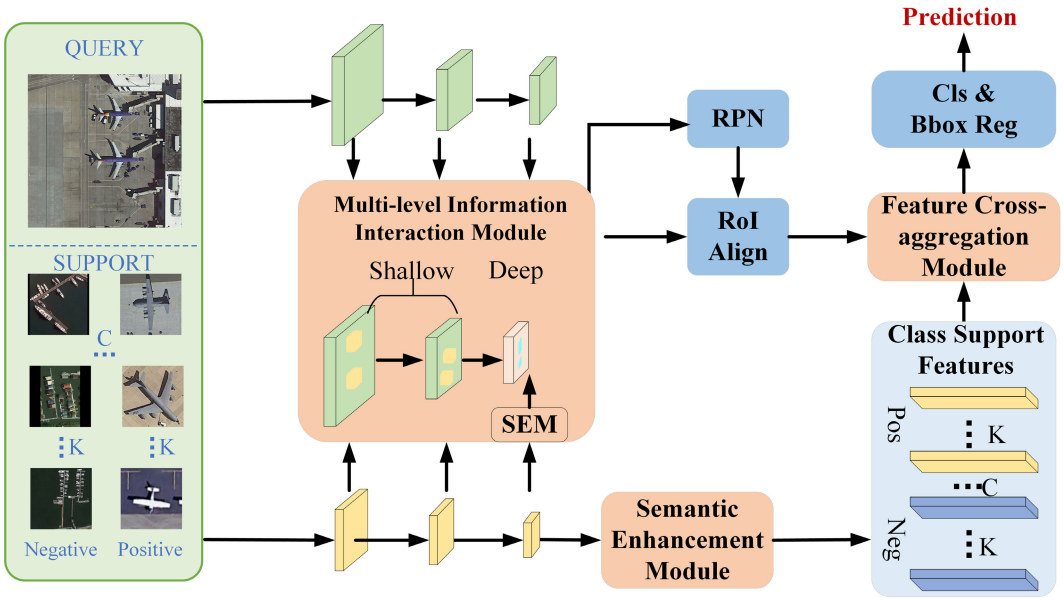


Fig. 2. Architecture of the proposed multilevel information interaction method. It consists of a parallel backbone to extract deep visual features for the input of a task, including a query image and C-way K-shot support images. The multilevel information interaction module performs two shallow interactions and a deep interaction. The SEM assists in selecting interactive information. Feature cross-aggregation module fuses positive and negative class support features into RoI visual features individually. A detection head is followed to perform classification and bbox refinement.

to train the detector for identifying objects that coexist in query and support images.

4) Comprehensive experiments have been conducted over several widely adopted benchmark datasets, which demonstrate the superiority of the proposed methods in both feature interaction and representation.

The remaining parts of the article are organized as follows. Section II describes our proposed MLII method in detail. Comprehensive experimental results are shown in Section III. The conclusion of this article is given in Section IV.

II. Methodology

In this section, we first formulate the problem and elaborate on the overall architecture. Then, the designed components are separately presented, including the multilevel feature interaction module, SEM, and FCM.

A. Problem Formulation

FSOD aims at proficiently extending the applicability of a detector that has been pretrained on base classes C_b with abundant data to novel classes C_n with only limited training samples where $C_n \cap C_b = \emptyset$. The datasets of base classes and novel classes are denoted as $D_b = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^{B}$ and $D_n = \{(\mathbf{x}_j, \mathbf{y}_j)\}_{j=1}^{N}$ ($N \ll B$), where $\mathbf{x}_i$ and $\mathbf{x}_j$ represent RSIs containing objects and $\mathbf{y}_i$ and $\mathbf{y}_j$ indicate the category of objects along with their bounding box (bbox) positions.

According to the metalearning paradigm, a series of tasks $\mathcal{T} = \{\mathcal{T}_b, \mathcal{T}_n\}$ are constructed from D_b and D_n, respectively. Within this framework, the tasks of base classes $\mathcal{T}_b = \{Q_i, S_i\}$ are exclusively sampled from D_b. These tasks encompass the query set Q_i and support set S_i, where the support set comprises c categories and k images for each category, adhering to the C-way K-shot principle. The knowledge detected query set from limited images C-way K-shot in support set is acquired during metatraining. FSOD based on metalearning transfers the knowledge learned from $\mathcal{T}_b$ into the tasks of novel classes $\mathcal{T}_n = \{Q_j, S_j\}$ sampling from D_n to improve detection performance on C_n. During the testing phase, all categories $C_t = C_b \cup C_n$ are detected concurrently simultaneously in the test datasets D_t.

B. Framework of the Proposed Method

By analyzing the aforementioned methods for creating support sets in RSI, as explained in Fig. 1, it is evident that support and query images exhibit inconsistencies in terms of size, scale, orientation, pose, and contextual information. These discrepancies lead to misalignment of features and attention bias in the two branches. If directly aligning or aggregating features from the deep layers of RSIs as done in previous methods [42], [43], [44], [45], [46], [47], there could be errors in similarity measurement, potentially misclassifying novel classes as base classes or background. This article introduces a novel FSOD framework with MLII for optical RSIs, which falls under the category of two-branch metalearning methods.

An overview of the proposed MLII is illustrated in Fig. 2 based on a typical two-stage object detection framework faster R-CNN. The parallel backbone network extracts deep visual features encompassing a query image and C-way K-shot support images. MLII is carried out through two shallow and deep interactions for two-branch images. The SEM is responsible for highlighting more representative key information. Subsequently, interacted features are used to generate proposals. RoI align is applied to extract region-of-interest (RoI) features by considering query features and proposals. The feature cross-aggregation module is employed to fuse RoI features with positive and negative class support features individually, including similarity measurements. The

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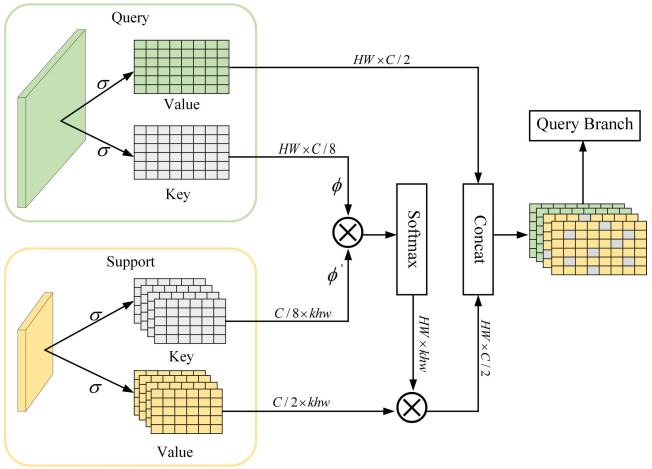


Fig. 3. Illustration of two shallow information interactions. Key maps are extracted to measure similarity as weight and value maps retain information for fusion.

detection head follows this step and executes classification and bbox refinement.

Notably, in RSIs, objects constitute a minor fraction of the overall image content. Hence, diverging from the traditional methodology, which supplements a standard three-channel RGB image with an additional one-channel binary mask outlining the object bboxs as inputs, as observed in [43] and [44], an alternative approach is pursued in this study. Specifically, object instance patches derived from the ground-truth bboxs are extracted as support images and resized to dimensions of 320 × 320 for input.

C. Multilevel Information Interaction Module

To address feature misalignment and attention bias between the support and query sets, the MLII module performs three levels of information interaction within the backbone of two branches. The two shallow interactions serve to align key information from two branches and integrate support key information into query features. The deep interaction adaptively learns the correspondence of region similarities between positive support features and query features.

1) Shallow Interactions of MLII: Shallow interactions adjust the attention bias with similarity measurements on key maps of two branches to decide which portion of the features to align. Value maps that sufficiently preserve distinct features of support are aligned and fused into the query value maps to update the query branch. In this process, shallow interactions serve to align features and adjust attention biases.

Given a task $\mathcal{T} = \{Q, S\}$, a single query image $Q \in \mathbb{R}^{1\times H\times W\times 3}$ and support images $S \in \mathbb{R}^{c\times k\times h\times w\times 3}$ are feed into a shared backbone network, which generates query and support features. These features then serve as the input for shallow interaction. To separately obtain key information for both support and query features, the two-branch features are encoded into pairs of key and value maps through dedicated deep encoders. The query encoder and support encoder share the same structure but do not share parameters.

As shown in Fig. 3, two sets of encoders map one or multiple features $\boldsymbol{F}_q \in \mathbb{R}^{1\times H\times W\times C}$, $\boldsymbol{F}_s \in \mathbb{R}^{c\times k\times h\times w\times C}$ into a key-value pair for each feature. Each encoder consists of two parallel 3 × 3 convolutional layers, which serve to key information while reducing the dimensionality to optimize computational efficiency. The key maps in shallow interaction are used to store class-specific distinctive features between query and support features, aiding in determining where to retrieve similar support values. Meanwhile, value maps maintain the original detailed information for the next level of feature exacted backbone. For query feature $\boldsymbol{F}_q$, the encoder carries out mapping to obtain the pair of key-value maps

$$\begin{aligned}(\boldsymbol{K}_q, \boldsymbol{V}_q) &= \sigma(\boldsymbol{F}_q)\\ \boldsymbol{K}_q &\in \mathbb{R}^{1\times H\times W\times C/8}\\ \boldsymbol{V}_q &\in \mathbb{R}^{1\times H\times W\times C/2}\end{aligned} \tag{1}$$

where σ is convolution operation, C is the feature dimension, H is the query height, and W is the query width of the input feature. For support features $\boldsymbol{F}_s$, the encoder outputs the pair of key-value maps for each of the features independently

$$\begin{aligned}(\boldsymbol{K}_s, \boldsymbol{V}_s) &= \sigma(\boldsymbol{F}_s)\\ \boldsymbol{K}_s &\in \mathbb{R}^{c\times k\times h\times w\times C/8}\\ \boldsymbol{V}_s &\in \mathbb{R}^{c\times k\times h\times w\times C/2}\end{aligned} \tag{2}$$

where $c \times k$ is the number of C-way K-shot, h is the support height, w is the support width, and C is the number of channels.

After obtaining the key-value maps of the query and support features, a two-branch information interaction is performed. Key maps of support and query features are performed similarity measurement as soft weights to determine which part of the information to interact with. The pixelwise similarity of key maps is formulated as

$$f\left(\boldsymbol{K}_q^i, \boldsymbol{K}_s^j\right) = \phi\left(\boldsymbol{K}_q^i\right)^{\mathrm{T}}\phi'\left(\boldsymbol{K}_s^j\right) \tag{3}$$

where $f(\boldsymbol{K}_q^i, \boldsymbol{K}_s^j) \in \mathbb{R}^{HW\times ckhw}$ denote all location similarity on key maps, and i and j are the indexes of HW and hw for the query and support. To enable the similarity function to dynamically learn during the training process, ϕ and ϕ' represent two independent linear transformations, and their parameters are learned through back propagation. The computed similarity is transformed into soft weights through softmax normalization

$$\boldsymbol{W}_{i,j} = \frac{\exp\left(f\left(\boldsymbol{K}_q^i, \boldsymbol{K}_s^j\right)\right)}{\sum_j \exp\left(f\left(\boldsymbol{K}_q^i, \boldsymbol{K}_s^j\right)\right)} \quad i = 1, 2, \ldots, HW; \quad j = 1, 2, \ldots, ckhw. \tag{4}$$

The sum of value maps on support features is emphasized by soft weights $\boldsymbol{W}$, and then, it is concatenated with the value map of the query feature. Hence, the output of shallow interaction is formulated as

$$\boldsymbol{F}_q^1 = \mathrm{concat}\left(\boldsymbol{V}_q, \boldsymbol{W} * \boldsymbol{V}_s\right) \tag{5}$$

where $*$ denotes the matrix multiplication, concat() represents the feature concatenation, and $\boldsymbol{W} * \boldsymbol{V}_s$ is the dynamic interactive information. The two shallow interactions are identical. $\boldsymbol{F}_q^1$ and $\boldsymbol{F}_q^2$ represent the updated query feature outputs, which are activated by the support features, where

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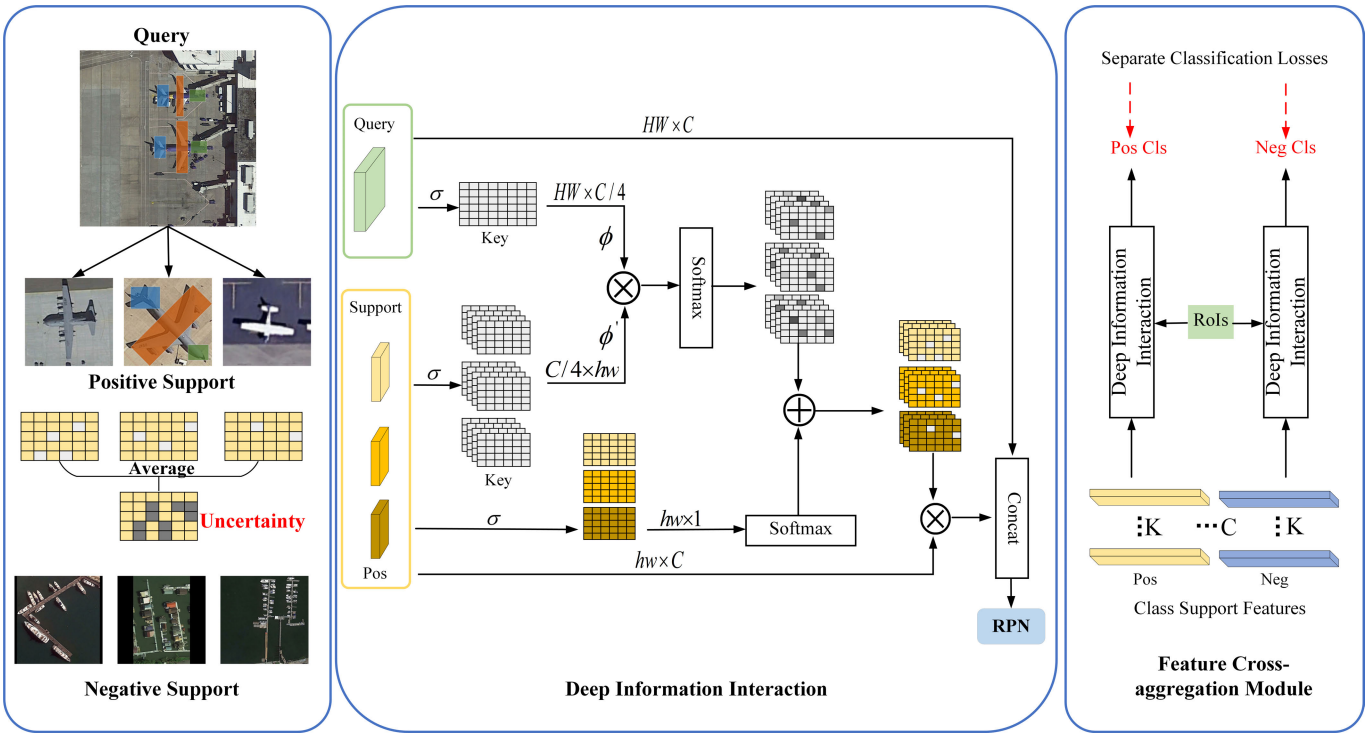


Fig. 4. Illustration of positive and negative support, deep information interaction, and FCM. Support images are divided into positive and negative according to whether it is coexisting objects between query and support samples. Due to the differences in appearance and orientation of positive support images, performing average pooling across multiple high-level features might cause uncertainty and blurred and confusing attention regions. The third information interaction on deep features is called deep information interaction. Class support features are from the last network block. The steps for cross-aggregating RoIs and class support features are the same as those for deep information interaction. The classifier determines whether the fused features belong to the current class.

exist a co-occurrence of object categories in both query and support images.

2) Deep Interaction of MLII: As illustrated in Fig. 4, support images are divided into positive and negative according to whether it is coexisting objects between query and support samples. Due to the random nature of the support images, positive support images exhibit certain differences in shape, appearance, texture, color, and background concerning the object in the query image. Performing average pooling [42], [43], [44] on high-level features from various positive supports may result in uncertainty [52], hindering the matching with the object in the query image. Deep interaction adaptively learns the correspondence between region similarities of the two branches.

Similarly, query features and K-shot positive features are encoded as key maps as follows:

$$\begin{aligned} \boldsymbol{K}_q &= \sigma\left(\boldsymbol{F}_q\right), \quad \boldsymbol{K}_q \in \mathbb{R}^{1\times HW\times C/4} \\ \boldsymbol{K}_{\mathrm{ps}} &= \sigma\left(\boldsymbol{F}_{\mathrm{ps}}\right), \quad \boldsymbol{K}_{\mathrm{ps}} \in \mathbb{R}^{k\times hw\times C/4}. \end{aligned} \tag{6}$$

Note that the channel count is doubled in the shallow interaction to retain more key information and merely encoded on positive features. The similarity scores between the query and positive support key maps are measured by

$$\boldsymbol{W}_d = \mathrm{softmax}\left(\phi\left(\boldsymbol{K}_q\right)^{\mathrm{T}}\phi'\left(\boldsymbol{K}_{\mathrm{ps}}\right)\right) \tag{7}$$

where $\boldsymbol{W}_d \in \mathbb{R}^{k\times HW\times hW}$ includes similarity score for each position. In addition, in deep interaction apart from the interaction between the two branches, more focus is given to self-distinguishable regions. Specifically, self-distinguishable maps are encoded as follows:

$$\boldsymbol{S}_{\mathrm{ps}}^k = \sigma\left(\boldsymbol{F}_{\mathrm{ps}}^k\right), \quad \boldsymbol{S}_{\mathrm{ps}}^k \in \mathbb{R}^{k\times H_s W_s\times 1} \tag{8}$$

where σ is a convolution operation. The final similarity added self-distinctive weight is formulated as

$$\boldsymbol{W}_d' = \boldsymbol{W}_d + \mathrm{softmax}\left(\boldsymbol{S}_{\mathrm{ps}}\right). \tag{9}$$

The output of deep interaction F_q^3 is formulated as

$$\boldsymbol{F}_q^3 = \mathrm{concat}\left(\boldsymbol{F}_q, \boldsymbol{W}_d' * \boldsymbol{F}_{\mathrm{ps}}\right) \tag{10}$$

where $\boldsymbol{F}_{\mathrm{ps}}$ is the sum of k positive support features. From the above calculation process, it can be observed that the deep interaction not only focuses on the similar interaction between the two branches but also highlights the unique characteristics of each positive support image. The rest of the support classes are called negative support, which do not participate in the interaction at this stage.

Due to the inconsistency in the appearance of support and query images (e.g., pose, size, and contextual information), unaligned features may lead to misclassifications. Equipped with the MLII module, it is possible to perform feature alignment at multiple levels, both in the shallow and deep layers. Pixels that coexist between the query set and support set will be highlighted to adjust attention biases.

D. Semantic Enhancement Module

The SEM aims to strengthen critical channels, emphasize highly discriminative regions, and suppress background and

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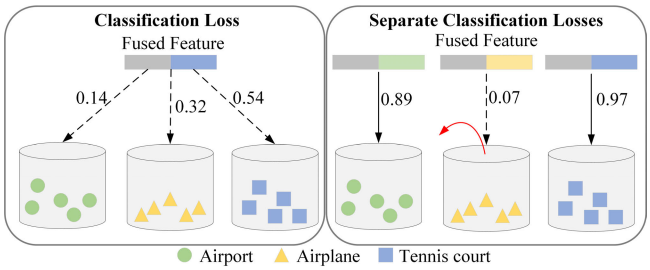


Fig. 5. Illustration of the ordinary classification loss and separate classification losses. During classification loss computation, the classifier tends to confuse the fused features into different support set categories. In the separated classification losses, it is easier for the classifier to determine whether the fused features belong to the same class as the current class support features. Red arrows indicate that they are not part of the current class.

confusable region features. In order to find narrow regions of high discrimination on the support features $\boldsymbol{F}_s = [p_1, p_2, \ldots, p_{hw}]$, each pixel feature $p_i \in \mathbb{R}^{1\times C}$ is first transformed into a value of pixel importance $\boldsymbol{M_g}(p_i)$. Specifically, a pixel feature p_i is transformed by a linear learnable matrix $\boldsymbol{M}_c \in \mathbb{R}^{C\times 1}$ and then normalized over all pixels, which is mathematically defined as

$$\boldsymbol{M_g}(p_i) = \text{softmax}(\boldsymbol{M}_c p_i) = \frac{\exp(\boldsymbol{M}_c p_i)}{\sum_{j\in hw}\exp(\boldsymbol{M}_c p_j)} \tag{11}$$

where p_i denotes the feature of ith the pixel and the softmax function is applied along the spatial dimension.

$\boldsymbol{M_g}(p_i)$ are then fused with the p_i to obtain the global channel weight $\boldsymbol{W}_g \in \mathbb{R}^{1\times C}$, which is formulated as

$$\boldsymbol{W}_g = \sum_i \boldsymbol{M}(p_i)\cdot p_i. \tag{12}$$

According to (12), $\boldsymbol{W}_g$ indicates the importance of each channel for $\boldsymbol{F}_s$. It is noticed that the global channel weight is mainly derived from discriminative narrow region features according to the importance of all spatial pixels. Since only a few narrow regions of support image make significant contributions, directly applying weights via $\boldsymbol{W}_g$ to the $\boldsymbol{F}_s$ would lose too much support information. Therefore, a softer approach is adopted to preserve support information as much as possible while highlighting discriminative features and suppressing background or confusable region features, which is defined as

$$\boldsymbol{F}'_s = \boldsymbol{F}_s + \alpha\cdot \text{LeakyReLU}\big(\boldsymbol{W}_g\big) \tag{13}$$

where α is a constant hyperparameter. The features $\boldsymbol{F}_s$ retain the support information and highlight essential channels by adding corrected weight, which is corrected by the nonlinear function Leaky ReLU. By adding the corrected weight, these discriminative narrow region features are enhanced or maintained, while features from irrelevant regions become blurred, thus exerting a suppressive effect. In addition, compared to the complex operations of the CBAM [58], SEM is more efficient because only $\boldsymbol{M}_c$ is the sole learnable weight. The emphasized channels assist MLII in extracting key maps and achieving more comprehensive feature representation.

E. Feature Cross-Aggregation Module

It has been validated that the primary factor affecting the performance of FSOD is classification error that misclassifies novel classes as confusable base classes or background [59]. Regression tasks can directly predict object positions based on ROI features without compromising FSOD performance. The FCM is designed to enhance classification performance for the classification task. The previous two-branch FSOD methods [42], [43], [44], [47] directly classify and calculate the classification loss on fused features. As illustrated in Fig. 5, the classifier allocates the fused features into the classes within the support set, which is difficult to distinguish and prone to classification confusion. To solve this problem, separate classification losses are introduced to train the model for more accurate classification.

The FCM aims to integrate class features with ROI features and adaptively learn the region correspondences between them. The separate fusions facilitate the design of separated classification losses, making it easier for the classifier to distinguish whether the ROI feature belongs to the category of the currently fused class feature. As shown in Fig. 4, the steps of feature cross-aggregation are the same as the deep interaction of MLII.

The fusion process of FCM is expressed as

$$\begin{aligned}\boldsymbol{F}_p &= \text{FCM}\big(\boldsymbol{Z}_r, \boldsymbol{Z}_p\big)\\ \boldsymbol{F}_n &= \text{FCM}(\boldsymbol{Z}_r, \boldsymbol{Z}_n)\end{aligned} \tag{14}$$

where $\boldsymbol{Z}_r$ is RoI features, and $\boldsymbol{Z}_p$ and $\boldsymbol{Z}_n$ denote positive and negative class support features. $\boldsymbol{F}_p$ and $\boldsymbol{F}_n$ are positive and negative fused features by FCM.

In addition, separate classification losses are introduced into the total loss to train model. The total loss of MLII is given as follows:

$$\mathcal{L} = \mathcal{L}_{\text{rpn}} + \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{pos}} + \mathcal{L}_{\text{neg}} \tag{15}$$

where $\mathcal{L}_{\text{rpn}}$ represents the classification and regression losses of RPN and $\mathcal{L}_{\text{reg}}$ denotes detection regression loss. The regression loss uses smooth L1 loss, and the classification loss applies cross-entropy loss. $\mathcal{L}_{\text{pos}}$ and $\mathcal{L}_{\text{neg}}$ denote separate positive and negative classification losses. The proposed separate classification losses are implemented, as shown in the following equation:

$$\begin{aligned}\mathcal{L}_{\text{pos}} &= \mathcal{L}_{\text{cls}}\big(t\big(\boldsymbol{F}_p\big), c\big(\boldsymbol{Z}_p\big)\big)\\ \mathcal{L}_{\text{reg}} &= \mathcal{L}_{\text{cls}}(t(\boldsymbol{F}_n), c(\boldsymbol{Z}_n))\end{aligned} \tag{16}$$

where $\mathcal{L}_{\text{cls}}$ is cross-entropy classification loss, and $c(\boldsymbol{Z}_p)$ and $c(\boldsymbol{Z}_n)$ are classes of positive and negative class support features. $t(\boldsymbol{F}_p)$ and $t(\boldsymbol{F}_n)$ denote the prediction of positive and negative fused features. During separate classification losses, the problem of ambiguity distinguishing different categories of fused features is effectively alleviated.

The complete metatraining and metatesting process of the proposed method is shown in Algorithm 1. In the metatraining of the proposed MLII method, shallow interactions adjust attention biases and align features to obtain more prominent features, thereby assisting in the final classification. Deep interaction and FCM learn the adaptive correspondence

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between support set and query set regions based on deep semantic features. Similar regions will be emphasized, aiding the query set in identifying objects. The ability to highlight similar region features gained during metatraining is also extended to the metatesting, helping the model to improve performance with few novel samples.

Algorithm 1 MLII Process Implementation

Input: A set of tasks $\mathcal{T} = \{Q, S\}$ which sampled from base classes dataset D_b and novel classes dataset D_n.

Output: Network model with parameters θ that performs well on novel classes.

```
Initialize the network parameters θ;
for stage in [meta-training, meta-testing] do
    for each task T_i do
        Extract features by backbone layer1;
        1st shallow interaction to acquire F_q^1;
        Extract features by backbone layer2;
        2nd shallow interaction to acquire F_q^2;
        Extract features by backbone layer3;
        SEM for positive support samples;
        Deep interaction to acquire F_q^3;
        Obtain region proposal and extract RoI features for F_q^3;
        FCM integrates positive and negative class support features into RoI features to generate fused features;
        Send fused features to the classification and regression head;
        Calculate separate classification losses L_pos, L_neg;
        Calculate the detection loss L_rpn and L_reg;
        Get the total training loss L in Eq. 15;
        update parameters θ ← θ − lr ∂L/∂θ;
    end for
end for
return result.
```

III. Experiments

This section demonstrates the experimental verification of the proposed algorithm on two public benchmark datasets of RSI object detection, including datasets, experimental settings, comparative experiments, and discussions of the proposed method.

A. Datasets

NWPU VHR-10 [60] contains 800 high-resolution satellite images, which are cropped from the Google Earth and Vaihingen datasets, and then manually annotated by experts. The dataset is divided into ten categories (aircraft, ships, storage tanks, baseball fields, tennis courts, basketball courts, ground runways, ports, bridges, and vehicles). The dataset contains a negative image set and a positive image set where the negative image set contains 150 images that do not contain any objects of a given object category and the positive image set contains 650 images; each image contains at least one target of interest.

DIOR [61], a large-scale, publicly available benchmark dataset in object detection RSIs, contains 23 463 images and 192 472 instances, covering 20 object classes including airplane, airport, baseball field, basketball court, bridge, chimney, dam, expressway service area, expressway toll station, harbor, golf course, ground track field, overpass, ship, stadium, storage tank, tennis court, train station, vehicle, and windmill. All images are of the size 800 × 800 pixels, and the spatial resolutions range from 0.5 to 30 m.

B. Experimental Settings

Two datasets are divided into base and novel datasets according to their classes to verify the performance of few-shot object detectors in remote sensing scenarios. As illustrated in Table I, DIOR and NWPU VHR-10 datasets designated five and three novel classes, respectively, and the remaining classes are considered base classes. In all experiments, the size of query images is set to 800 × 800, corresponding to the pictures of DIOR and NWPU datasets. The support images contain only one target, cropped based on the ground-truth bboxs from the DIOR and NWPU datasets and uniformly resized to 320 × 320. During the metatraining stage, both the support set and the query set consist of base classes. In each epoch, the support set comprises randomly sampled few-shot samples of C-way K-shot. During the metatesting stage, both the support and query sets are derived from a small balanced dataset containing all classes, including base and novel classes. The support set for base classes consists of randomly sampled K-shot samples, while, for novel classes, it consists of the only available K-shot samples. K is set to 3, 5, 10, and 20, respectively.

To demonstrate the effectiveness of the proposed method, some state-of-the-art methods are compared with ours. All experiments are entirely consistent in parameters, such as equipment, the number of iterations, decay rate, and anchor size, to ensure the fairness of the comparison to the greatest extent. Different random seeds are used to select support images in each experiment to reduce the effect of varying support images on the experimental results. Experiments are carried out with ten runs to obtain the average value.

Experiments for all methods adopt ResNet-101 pretrained on ImageNet as the base detector's backbone with anchor rations set at 0.25, 0.5, 1, 2, and 4, respectively. The models are optimized by the stochastic gradient descent (SGD) method. The base learning rate is 0.002, the warm-up distill is 200, and the batch size is 4. For base classes, the model is trained for 20 epochs and ten epochs on the balanced dataset of novel and base classes, respectively. The learning rate decay step is 12. In addition, all experiments are conducted on NVIDIA GeForce RTX 3090 GPU.

C. Comparisons With Sate-of-the-Art Methods

1) Experiments on DIOR: The proposed model is compared with state-of-the-art methods on the DIOR dataset, including three, five, ten, and 20 shots. The experimental results are demonstrated in Table II. From the experimental results, it can be observed that the detection performance of almost

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TABLE I
NOVEL AND BASE CLASSES IN EACH DATASET FOR FEW-SHOT SCENARIOS

Dataset	Novel Classes	Base Classes
DIOR	windmill, airplane, tennis court, train station, baseball field	dam, stadium, ground track field, basketball court, expressway toll station, storage tank, airport, vehicle, golf field, harbor, ship, expressway service area, overpass, bridge, chimney
NWPU VHR-10	airplane, baseball diamond, tennis court	ship, storage tank, harbor, bridge, vehicle, ground track field

TABLE II
FEW-SHOT OBJECT DETECTION RESULTS ON THE DIOR DATASET

Shot	Methods	Novel Classes					mAP
		windmill	airplane	tennis court	train station	baseball field	
3	Meta RCNN [44]	9.1	15.1	31.8	7.1	5.2	13.7
	FsDetView [43]	9.1	9.7	22.8	6.9	4.3	10.6
	DAnA [52]	0.3	1.9	19.4	14.1	19.3	11.0
	FSCE [40]	9.1	9.1	6.6	9.9	27.3	12.4
	DCFS [41]	9.1	11.7	23.6	3.5	37.0	17.0
	MFDC [38]	11.2	10.1	35.2	7.0	40.1	20.8
	MLII-FSOD(ours)	10.4	19.2	22.5	15.6	36.8	**20.9**
5	Meta RCNN [44]	9.1	17.0	33.2	8.9	7.2	15.1
	FsDetView [43]	9.1	22.3	26.2	5.2	2.3	13.0
	DAnA [52]	2.4	11.4	26.8	12.5	36.5	17.9
	FSCE [40]	15.6	9.1	40.6	10.2	41.9	23.5
	DCFS [41]	13.1	15.8	39.5	8.5	43.5	24.1
	MFDC [38]	14.8	13.4	40.5	10.5	45.5	24.9
	MLII-FSOD(ours)	12.8	21.9	30.2	23.4	38.4	**25.3**
10	Meta RCNN [44]	11.4	33.5	32.4	14.0	3.5	18.9
	FsDetView [43]	11.2	20.4	25.6	7.0	6.1	14.1
	DAnA [52]	6.5	21.1	27.9	15.9	25.5	19.4
	FSCE [40]	27.5	9.1	38.8	14.7	47.2	27.5
	DCFS [41]	15.9	18.9	48.8	14.4	46.9	29.0
	MFDC [38]	15.6	17.8	47.7	16.4	49.0	29.3
	MLII-FSOD(ours)	27.7	20.5	38.4	26.2	47.8	**32.1**
20	Meta RCNN [44]	10.2	34.5	35.4	13.7	8.5	20.5
	FsDetView [43]	16.2	31.7	25.8	11.1	6.1	18.2
	DAnA [52]	10.4	14.3	28.6	16.5	38.5	21.7
	FSCE [40]	34.1	20.1	49.1	13.5	51.1	33.6
	DCFS [41]	14.0	19.2	49.7	16.7	49.7	29.9
	MFDC [38]	16.0	18.3	48.3	16.6	45.9	29.0
	MLII-FSOD(ours)	35.2	34.2	39.1	26.5	48.2	**36.6**

all methods gradually improves with the number increase of novel class samples. Compared with these state-of-the-art FSOD methods, the proposed model (MLII) achieves the highest performance on mAPs for novel classes reaching 20.9%, 25.3%, 32.1%, and 36.6% on three, five, ten, and 20 shots respectively. The results prove the effectiveness of our proposed method of FSOD in remote sensing.

In addition, the results are visualized in Fig. 6 for the proposed method and without the MLII module on novel classes. It can be seen that the method without the MLII module can locate the objects belonging to novel classes because a simple fusion of features after RoI align still provides support information. However, due to the insufficient shallow features to supplement, these novel objects are, thus, misclassified as base class or background. For example, small airplanes or targets on the edge of the RSI are missed. When multiple class objects are in the same image, some tennis courts with unobvious appearances are recognized as backgrounds. The proposed method strengthens the feature representation of objects by introducing MLII, which makes it easier to discover objects and alleviates classification confusion effectively.

2) Experiments on NWPU VHR-10: The experimental results on NWPU VHR-10 are presented in Table III. The novel classes airplane, baseball diamond, and tennis court are abbreviated as AL, BD, and TC, respectively. It is observed that the performances of these compared FSOD methods on the NWPU VHR-10 dataset are sorted differently from the DIOR dataset, i.e., the performance of FSCE methods is higher than other metalearning-based and fine-tuning-based methods. Due to the multiscale strategy, FSCE greatly improves the performance on the NWPU VHR-10 dataset, but

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Fig. 6. Visualization comparison between the proposed method MLII (line 2 and line 4) and without MLII (line 1 and line 3) on the DIOR dataset of the ten-shot setting. Only the bboxs with a confidence greater than 0.3 are visualized. The objects circled by the red ellipse are misidentified as background or misclassified.

this advantage is not obvious on the DIOR dataset. Compared to these methods, the proposed method achieves substantial performance gains under all shot settings, and as the number of shots increases, the performance of the model improves rapidly. For example, under the five-shot setting, the mAP of the proposed method is 26.6% higher than the highest value in the comparative method.

As shown in Fig. 7, the visualization results are compared between the proposed method (MLII) and without MLII on the NWPU VHR-10 dataset under the ten-shot setting. The method without MLII suffers from many misclassifications, i.e., misclassifying novel classes as base classes or backgrounds. For example, the method without MLII misclassifies vehicles and backgrounds as airplanes, tennis courts, and basketball courts as background. However, the proposed method can extract the distinctive features of categories by MLII and improve the performance of novel classes.

D. Ablation Studies

To demonstrate the effectiveness of the proposed method MLII and its information interaction, sufficient ablation experiments are conducted on the NWPU VHR-10 dataset.

1) Ablation Study for Each Information Interaction: Table IV shows the effectiveness of each information interaction in NWPU VHR-10. Shallow1 and Shallow2 denote two shallow interactions of MLII, and deep indicates deep interaction of MLII. By introducing the MLII module, the detection performance for novel classes is significantly improved, and it has reached 41%, 56.1%, 64.8%, and 70.4% under the settings of three-shot, five-shot, ten-shot, and 20-shot, respectively.

The experimental results indicate that each form of information interaction contributes to the enhancement of the performance of novel classes (lines 4–6). The results

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TABLE III
FEW-SHOT OBJECT DETECTION RESULTS ON THE NWPU VHR-10 DATASET

Methods	Shot															
	3				5				10				20			
	AL	BD	TC	mAP	AL	BD	TC	mAP	AL	BD	TC	mAP	AL	BD	TC	mAP
Meta RCNN [44]	6.9	7.2	5.8	6.6	12.3	21.7	1.9	12.0	11.5	22.4	12.5	15.5	13.6	24.6	16.8	18.3
FsDetView [43]	9.1	19.7	6.8	11.9	14.3	26.8	10.8	17.3	16.1	20.1	16.4	17.5	18.8	30.2	12.4	20.5
DAnA [52]	16.2	10.3	13.5	13.3	15.6	14.5	12.9	14.3	27.7	24.4	19.6	23.9	28.8	27.0	23.0	26.3
DCFS [41]	18.4	38.2	17.3	25.0	34.6	70.2	18.5	41.1	52.8	**80.7**	36.2	56.6	44.5	78.6	34.4	52.5
MFDC [38]	31.8	36.9	17.7	28.8	50.2	57.3	21.6	43.0	62.4	69.1	26.8	52.7	55.4	71.7	34.4	53.8
FSCE [40]	29.1	51.6	25.0	35.2	42.8	66.0	26.3	45.1	60.7	76.8	33.0	56.8	72.3	**81.9**	42.9	65.7
MLII-FSOD(ours)	**42.3**	**57.5**	**30.2**	**43.3**	**52.4**	**71.3**	**47.7**	**57.1**	**74.2**	72.1	**52.0**	**66.1**	**88.7**	72.2	**53.3**	**71.4**

Abbreviation of novel classes - AL: airplane; BD: baseball diamond; TC: tennis court.

TABLE IV
ABLATION STUDY FOR EACH INFORMATION INTERACTION

MLII			Shot			
Shallow1	Shallow2	Deep	3	5	10	20
✓			23.9	40.6	46.6	48.6
	✓		24.6	43.0	48.3	49.3
		✓	36.2	50.4	59.0	60.6
✓	✓		31.9	47.2	54.3	57.9
✓		✓	36.9	51.8	59.7	61.3
	✓	✓	38.4	54.9	62.2	64.5
✓	✓	✓	**41.0**	**56.1**	**64.8**	**70.4**

TABLE V
ABLATION STUDY FOR EACH COMPONENT

Component			Shot			
MLII	SEM	FCM (separate losses)	3	5	10	20
✓	✓		37.9	54.2	64.3	67.9
✓		✓	39.2	55.8	65.7	70.3
	✓	✓	38.1	53.0	62.5	67.5
✓	✓	✓	**43.3**	**57.1**	**66.1**	**71.4**

from utilizing deep interaction in isolation (line 3) surpass the performance achieved when both shallow interactions are employed simultaneously (line 4). This suggests that deep interactions play a predominant role. In cases where the model exclusively incorporates two shallow interactions, the overall performance improvement is not significant. It is only after achieving semantic alignment with deep-level features that a significant boost in the overall model performance is observed, highlighting the auxiliary role played by shallow interactions.

2) Ablation Study for Each Component: Eliminating the FCM and MLII is impossible due to the two-branch fusion process required before detection. The SEM primarily serves as a module for learning key information and enhancing representations and cannot be used independently without MLII and FCM modules. Detailed ablation analysis of the MLII component has been discussed in the previous section and will not be reiterated. The validation of the effectiveness of each component is presented in Table V. As shown in the results, each component improves the model's performance. Notably, more excellent enhancement is observed when MLII and FCM are employed simultaneously, indicating that the SEM module is a practical auxiliary component. Optimum performance is achieved when all three components are used together.

E. Discussion

1) Discussion on Information Interaction Methods: As shown in Fig. 8, different information interaction methods are applied in the discussion to eliminate the two-branch unaligned. The four information interaction methods other than the proposed MLII are explained in detail below.

BI + Concat (Support Key): The support key features are obtained by convolution support features, and then, bilinear interpolation (BI) is used to expand the key feature spatial size to be consistent with the query feature. The expanded feature is directly concatenated with the query feature, as shown in Fig. 8(a).

BI + Concat (Support Key and Query Key): The support and query key features are obtained by convolution, and then, BI is used to expand the key feature spatial size to be consistent with the query feature. The expanded features are first concatenated with the query key features and then concatenated with the original query features, as shown in Fig. 8(b).

BI + Addition + Concat (Support Key and Query Key): Support key features and query key features are obtained by convolution, and then, BI is used to extend the spatial size of the support key features to be consistent with the query features. The extended support key features and query key features are summed and then concatenated with the original query features, as shown in Fig. 8(c).

BI + Multiplication + Concat (Support Key and Query Key): The support key features and query key features are obtained by convolution, and then, the spatial size of the support key features is extended using BI to make it consistent with the query features. The query key features are multiplied by the weights of the extended support key features transformed by softmax and then concatenated with the original query features, as shown in Fig. 8(d).

The experimental results are presented in Table VI. Experimental results show that simply concatenating support

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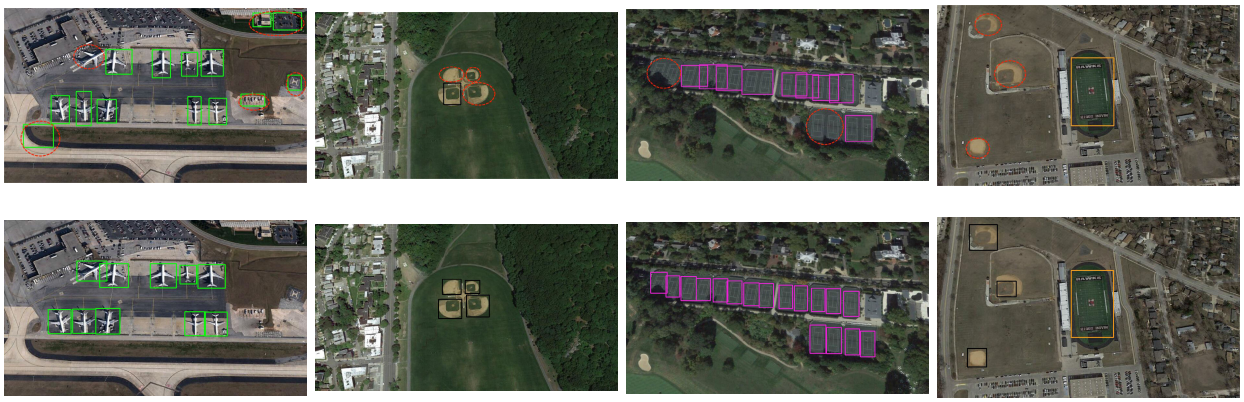


Fig. 7. Visualization comparison between the MLII (line 2) and without MLII (line 1) on the NWPU VHR-10 dataset of ten-shot setting. Only the bboxs with a confidence greater than 0.3 are visualized. The objects circled by the red ellipse are misidentified as background or misclassified.

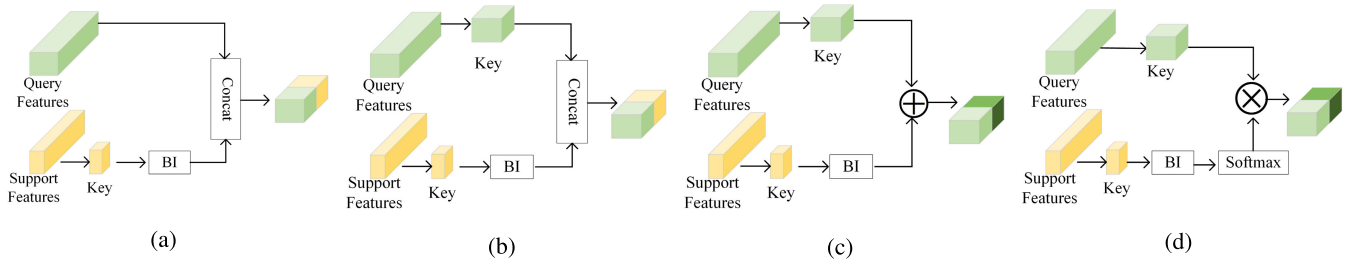


Fig. 8. Four different information interaction methods for two branches to eliminate unaligned. BI: bilinear interpolation; $\oplus$: addition; and $\otimes$: multiplication. (a) BI + Concat (support key). (b) BI + Concat (support key and query key). (c) BI + Addition + Concat (support key and query key). (d) BI + Multiplication + Concat (support key and query key).

TABLE VI
DISCUSSION ON INFORMATION INTERACTION METHODS

method	3-SHOT	5-SHOT	10-SHOT
BI + Concat (Support Key)	31.1	37.2	50.2
BI + Concat (Support Key and Query Key)	31.6	37.7	50.4
BI + Addition + Concat (Support Key and Query Key)	27.5	33.8	48.5
BI + Multiplication + Concat (Support Key and Query Key)	34.9	45.7	52.1
Ours	**42.0**	**57.1**	**66.1**

TABLE VII
INFLUENCE OF THE NUMBER OF KEYS AND VALUES

KEY	VALUE	3-SHOT	10-SHOT
Key = 1/16	Value = 1/2	34.1	52.9
Key = 1/8	Value = 1/2	43.3	**66.1**
Key = 1/4	Value = 1/2	38.4	50.5
Key = 1/16	Value = 3/4	34.2	51.0
Key = 1/8	Value = 3/4	**43.4**	63.5
Key = 1/4	Value = 3/4	38.3	51.8

key features [e.g., Fig. 8(a) and (b)] does not eliminate two-branch misalignment, resulting in insignificant performance improvement. As shown in Fig. 8(c), directly adding support and query key features may cause a puzzle between the two branch information, resulting in slight performance degradation. Through an attention mechanism, as shown in Fig. 8(d), the support key features are used as a weight to adjust the attention area of the query key features. However, due to the inconsistency of the feature scale, there is a bias in the attention area, leading to the vital position not being paid enough attention. Furthermore, the proposed MLII achieves the highest performance under all shot settings because it corrects feature misalignment by similarity measures on key features and the retention of the useful information on value features, which readjusted the attention position to be consistent with the query feature. Thus, the proposed MLII can be applied to accurately emphasize the relevant information representation in query features and preserve the discriminative information on support features.

2) Influence of the Number of Keys and Values: Different keys and values are discussed in detail to verify the impact of different numbers of keys and values on the performance of the MLII model. The key maps are respectively set as 1/4, 1/8, and 1/16 of the original features. In order to retain enough information, the value maps are respectively set as 1/2 and 3/4 of the original features. The experimental results are displayed in Table VII. It is observed that the best performance is achieved when key = 1/8 and value = 1/2. The performance does not vary much when key = 1/8 and value = 3/4. As the number of keys decreases, the information extracted by the key maps decreases, so the performance of the proposed model gradually decreases. More key maps will scatter key information, thus making learning more difficult.

3) Discussion on Semantic Enhancement Methods: A series of comparative experiments are conducted to verify the effectiveness of the semantic enhancement method. The experiment of the SEM is shown in Table VIII. The original support features $\boldsymbol{F}_s$ are directly used to extract the key maps without performing semantic enhancement operations. For semantic enhancement, $\boldsymbol{W}_g$ is used as a weight to directly perform channel weighting with the original support features $\boldsymbol{F}_s$. Compared with using the original support features,

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TABLE VIII
DISCUSSION ON SEMANTIC ENHANCEMENT METHODS

Semantic Enhancement Method	3-SHOT	5-SHOT	10-SHOT
$\boldsymbol{F}_s$	40.0	49.2	55.3
$\boldsymbol{W}_g$	29.2	34.3	39.8
CBAM [59]	40.6	49.7	55.8
SEM	**43.3**	**57.1**	**66.1**

TABLE IX
DISCUSSION ON THE LOCATION OF SEM

The location of SEM			Shot			
Shallow1	Shallow2	Deep	3	5	10	20
✓			14.3	15.8	23.9	26.3
	✓		20.1	25.2	27.3	29.5
		✓	**41.0**	**56.1**	**64.8**	**70.4**

the way directly weighted channels lose too much support information, resulting in performance degradation. Compared with the weighted way, the SEM performance is improved by more than 43%. Furthermore, the performance is also improved by 19.5% compared with directly using the original support features under a ten-shot setting. CBAM enhances the effectiveness of the original features $\boldsymbol{F}_s$ by performing spatial attention and channel attention. However, it does not achieve the same level of performance as SEM, and its operations are more complex. In conclusion, the experimental results demonstrate the superiority of the proposed SEM over the comparison methods.

4) Discussion on the Location of SEM: The SEM is added to different stages of information interaction, as shown in Table IX. Shallow1 and Shallow2 denote two shallow interactions of MLII, and deep indicates deep interaction of MLII. It can be observed that when the SEM module is added to shallow interactions, the performance noticeably declines. Theoretically, the global channel weight of SEM is derived from important spatial pixels. In shallow features, the spatial size is larger than in deep features. Thus, important regions might be dispersed. This implies that the obtained global channel weight is composed of multiple scattered regions of pixels. When added to the original features, this may lead to confusion, whereas, in deep features, the global channel weight is mainly derived from discriminative narrow region features considering the importance of all spatial pixels. After adding the global channel weight, these narrow region features are enhanced or maintained, while features from irrelevant regions become blurred, thus exerting a suppressive effect.

5) Hyperparameter Discussion: The discussion of hyperparameter α in (13) is conducted on the NWPU VHR-10 dataset under the five-shot setting to determine the appropriate proportion, as shown in Fig. 9. When $\alpha = 0$, the module degenerates to directly use the support feature to extract the key maps. When $\alpha = 0.1$, the proposed method achieves optimal performance by preserving adequate support information and emphasizing critical features. As α exceeds 0.1, performance gradually degrades. $\boldsymbol{W}_g$ becomes more influential, meaning incline to narrow region features, resulting in a loss of more support information. However, the SEM still achieves favorable performance as long as $\alpha \leq 0.7$, which proves that the SEM can assist the MLII module in emphasizing the key maps in advance without losing the support information.



Fig. 9. Hyperparameter α analysis experiment.

IV. CONCLUSION

In this article, we show how support sets are constructed in RSIs and find that there are inconsistencies between query sets and support sets in terms of scale, orientation, background, pose, and so on. These inconsistencies lead to feature misalignment and attention bias, which result in performance degradation. Thus, a novel FSOD framework with MLII is proposed to solve the problem of feature misalignment and attention bias. The MLII module includes two shallow interactions and a deep interaction. The shallow interactions aim to align the key information and adjust the attention to emphasize mutual elements more strongly. The deep interaction adaptively learns the correspondence between region similarities of the two branches. In addition, the SEM enhances essential channels, and the FCM fuses class support features with RoI features. Separate classification losses are introduced to resolve the ambiguity in distinguishing between different classes. Extensive experiments demonstrate that the proposed method outperforms several state-of-the-art few-shot object detectors over commonly used benchmark datasets DIOR and NWPU-10. The limitation of the proposed method is the extra computational effort associated with multiple similarity measurements. In future work, we will explore how to better interact with the network structure of two-branch information without increasing the computational burden.

REFERENCES

[1] S. Mei, Y. Geng, J. Hou, and Q. Du, "Learning hyperspectral images from RGB images via a coarse-to-fine CNN," *Sci. China Inf. Sci.*, vol. 65, no. 5, pp. 1–14, May 2022.

[2] J. Li, P. Tian, R. Song, H. Xu, Y. Li, and Q. Du, "PCViT: A pyramid convolutional vision transformer detector for object detection in remote-sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–15, 2024, Art. no. 5608115, doi: 10.1109/TGRS.2024.3360456.

[3] Y. Wang et al., "Remote sensing image super-resolution and object detection: Benchmark and state of the art," *Expert Syst. Appl.*, vol. 197, Jul. 2022, Art. no. 116793.

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[4] W. Ma et al., "Feature split–merge–enhancement network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022, Art. no. 5616217, doi: 10.1109/TGRS.2022.3140856.

[5] C. Xu, C. Li, Z. Cui, T. Zhang, and J. Yang, "Hierarchical semantic propagation for object detection in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4353–4364, Jun. 2020.

[6] Y. Yu, H. Guan, D. Li, T. Gu, E. Tang, and A. Li, "Orientation guided anchoring for geospatial object detection from remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 160, pp. 67–82, Feb. 2020.

[7] Z. Han, Z. Zhang, S. Zhang, G. Zhang, and S. Mei, "Aerial visible-to-infrared image translation: Dataset, evaluation, and baseline," *J. Remote Sens.*, vol. 3, p. 96, Jan. 2023.

[8] S. Mei, R. Jiang, M. Ma, and C. Song, "Rotation-invariant feature learning via convolutional neural network with cyclic polar coordinates convolutional layer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–13, 2023, Art. no. 5600713, doi: 10.1109/TGRS.2022.3233726.

[9] J. Lian, S. Mei, S. Zhang, and M. Ma, "Benchmarking adversarial patch against aerial detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022, Art. no. 5634616, doi: 10.1109/TGRS.2022.3225306.

[10] J. Lian, X. Wang, Y. Su, M. Ma, and S. Mei, "CBA: Contextual background attack against optical aerial detection in the physical world," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–16, 2023, Art. no. 5606616, doi: 10.1109/TGRS.2023.3264839.

[11] S. Mei, J. Lian, X. Wang, Y. Su, M. Ma, and L.-P. Chau, "A comprehensive study on the robustness of image classification and object detection in remote sensing: Surveying and benchmarking," 2023, *arXiv:2306.12111*.

[12] J. Li, S. Du, R. Song, Y. Li, and Q. Du, "Progressive spatial information-guided deep aggregation convolutional network for hyperspectral spectral super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, 2024.

[13] P. Kaul, W. Xie, and A. Zisserman, "Label, verify, correct: A simple few shot object detection method," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 14237–14247.

[14] T. Zhang, X. Zhang, P. Zhu, X. Jia, X. Tang, and L. Jiao, "Generalized few-shot object detection in remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 195, pp. 353–364, Jan. 2023.

[15] S. Lin, K. Wang, X. Zeng, and R. Zhao, "Explore the power of synthetic data on few-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 638–647.

[16] Q. Liu, J. Peng, G. Zhang, W. Sun, and Q. Du, "Deep contrastive learning network for small-sample hyperspectral image classification," *J. Remote Sens.*, vol. 3, p. 0025, Jan. 2023.

[17] C. Zhou, J. Zhang, J. Liu, C. Zhang, G. Shi, and J. Hu, "Bayesian transfer learning for object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7705–7719, Nov. 2020.

[18] A. Elshamli, G. W. Taylor, and S. Areibi, "Multisource domain adaptation for remote sensing using deep neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3328–3340, May 2020.

[19] Y. Koga, H. Miyazaki, and R. Shibasaki, "A method for vehicle detection in high-resolution satellite images that uses a region-based object detector and unsupervised domain adaptation," *Remote Sens.*, vol. 12, no. 3, p. 575, Feb. 2020.

[20] G. Cheng, C. Yang, X. Yao, L. Guo, and J. Han, "When deep learning meets metric learning: Remote sensing image scene classification via learning discriminative CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2811–2821, May 2018.

[21] H. Li et al., "RS-MetaNet: Deep metametric learning for few-shot remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6983–6994, Aug. 2021.

[22] M. Werther et al., "Meta-classification of remote sensing reflectance to estimate trophic status of inland and nearshore waters," *ISPRS J. Photogramm. Remote Sens.*, vol. 176, pp. 109–126, Jun. 2021.

[23] J. Li, Z. Zhang, R. Song, Y. Li, and Q. Du, "SCFormer: Spectral coordinate transformer for cross-domain few-shot hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 33, pp. 840–855, 2024.

[24] S. X. Hu, D. Li, J. Stühmer, M. Kim, and T. M. Hospedales, "Pushing the limits of simple pipelines for few-shot learning: External data and fine-tuning make a difference," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 9068–9077.

[25] L. Tunstall et al., "Efficient few-shot learning without prompts," 2022, *arXiv:2209.11055*.

[26] J. Ma, H. Xie, G. Han, S.-F. Chang, A. Galstyan, and W. Abd-Almageed, "Partner-assisted learning for few-shot image classification," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10553–10562.

[27] Z. Ji, L. Hou, X. Wang, G. Wang, and Y. Pang, "Dual contrastive network for few-shot remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–12, 2023, Art. no. 5605312, doi: 10.1109/TGRS.2023.3260121.

[28] C. Lang, J. Wang, G. Cheng, B. Tu, and J. Han, "Progressive parsing and commonality distillation for few-shot remote sensing segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–10, 2023, Art. no. 5613610, doi: 10.1109/TGRS.2023.3286183.

[29] X. Li, J. Deng, and Y. Fang, "Few-shot object detection on remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2021, Art. no. 5601614.

[30] W. Zhang and Y.-X. Wang, "Hallucination improves few-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13008–13017.

[31] C. Zhu, F. Chen, U. Ahmed, Z. Shen, and M. Savvides, "Semantic relation reasoning for shot-stable few-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 8782–8791.

[32] Y. Cao et al., "Few-shot object detection via association and discrimination," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 16570–16581.

[33] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[34] C.-Y. Wang, A. Bochkovskiy, and H.-Y.-M. Liao, "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7464–7475.

[35] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," 2018, *arXiv:1804.02767*.

[36] W. Liu et al., "SSD: Single shot MultiBox detector," in *Computer Vision—ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part I 14*. Springer, 2016, pp. 21–37.

[37] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 734–750.

[38] S. Wu, W. Pei, D. Mei, F. Chen, J. Tian, and G. Lu, "Multi-faceted distillation of base-novel commonality for few-shot object detection," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 578–594.

[39] Z. Yang, C. Zhang, R. Li, Y. Xu, and G. Lin, "Efficient few-shot object detection via knowledge inheritance," *IEEE Trans. Image Process.*, vol. 32, pp. 321–334, 2023.

[40] B. Sun, B. Li, S. Cai, Y. Yuan, and C. Zhang, "FSCE: Few-shot object detection via contrastive proposal encoding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 7352–7362.

[41] B. B. Gao et al., "Decoupling classifier for boosting few-shot object detection and instance segmentation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 18640–18652.

[42] G. Han, S. Huang, J. Ma, Y. He, and S.-F. Chang, "Meta faster R-CNN: Towards accurate few-shot object detection with attentive feature alignment," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 780–789.

[43] Y. Xiao, V. Lepetit, and R. Marlet, "Few-shot object detection and viewpoint estimation for objects in the wild," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 3, pp. 3090–3106, Mar. 2023.

[44] X. Yan, Z. Chen, A. Xu, X. Wang, X. Liang, and L. Lin, "Meta R-CNN: Towards general solver for instance-level low-shot learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9576–9585.

[45] A. Li and Z. Li, "Transformation invariant few-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 3094–3102.

[46] B. Li, B. Yang, C. Liu, F. Liu, R. Ji, and Q. Ye, "Beyond max-margin: Class margin equilibrium for few-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 7363–7372.

[47] G. Han, J. Ma, S. Huang, L. Chen, and S.-F. Chang, "Few-shot object detection with fully cross-transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5321–5330.

[48] B. Kang, Z. Liu, X. Wang, F. Yu, J. Feng, and T. Darrell, "Few-shot object detection via feature reweighting," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Jun. 2019, pp. 8420–8429.

[49] Q. Fan, W. Zhuo, C. Tang, and Y. Tai, "Few-shot object detection with attention-RPN and multi-relation detector," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4012–4021.

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[50] G. Han, Y. He, S. Huang, J. Ma, and S.-F. Chang, "Query adaptive few-shot object detection with heterogeneous graph convolutional networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3263–3272.
[51] D.-J. Chen, H.-Y. Hsieh, and T.-L. Liu, "Adaptive image transformer for one-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 12247–12256.
[52] T.-I. Chen et al., "Dual-awareness attention for few-shot object detection," *IEEE Trans. Multimedia*, vol. 25, pp. 291–301, 2023.
[53] C. Doersch, A. Gupta, and A. Zisserman, "CrosstransFormers: Spatially-aware few-shot transfer," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 21981–21993.
[54] T.-I. Hsieh, Y.-C. Lo, and H.-T. Chen, "One-shot object detection with co-attention and co-excitation," 2019, pp. 2702–2711.
[55] G. Cheng et al., "Prototype-CNN for few-shot object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–10, 2022, Art. no. 5604610, doi: 10.1109/TGRS.2021.3078507.
[56] Z. Xiao, J. Qi, W. Xue, and P. Zhong, "Few-shot object detection with self-adaptive attention network for remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4854–4865, 2021.
[57] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional networks," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 818–833.
[58] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.
[59] X. Lu et al., "Few-shot object detection in aerial imagery guided by text-modal knowledge," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5604719.
[60] G. Cheng, J. Han, P. Zhou, and L. Guo, "Multi-class geospatial object detection and geographic image classification based on collection of part detectors," *ISPRS J. Photogramm. Remote Sens.*, vol. 98, pp. 119–132, Dec. 2014.
[61] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020, doi: 10.1016/j.isprsjprs.2019.11.023.



Lefan Wang (Graduate Student Member, IEEE) received the B.S. degree in electronic commerce from Liaocheng University, Liaocheng, China, in 2017, and the M.S. degree in computer technology from Northwestern Polytechnical University, Xi'an, China, in 2020, where she is currently pursuing the Ph.D. degree in information and communication engineering.

Her research interests include remote sensing and image processing.



Shaohui Mei (Senior Member, IEEE) received the B.S. degree in electronics and information engineering and the Ph.D. degree in signal and information processing from Northwestern Polytechnical University, Xi'an, China, in 2005 and 2011, respectively.

He was a Visiting Student with The University of Sydney, Camperdown, NSW, Australia, from October 2007 to October 2008. He is currently a Tenured Professor with the School of Electronics and Information, Northwestern Polytechnical University. His research interests include hyperspectral remote sensing image processing and applications, intelligent signal and information acquisition and processing, video processing, and pattern recognition.

Dr. Mei received the First Prize of Natural Science Award of Shaanxi Province in 2022, the Excellent Doctoral Dissertation Award of Shaanxi Province in 2014, and the First Place Award certificate for both "Hyperspectral 1004 Object Tracking Challenge" and "Cross-City Multimodal Semantic Segmentation Challenge" in WHISPERS 2023. He is serving as an Associate Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, a Guest Editor for *Remote Sensing*, and a reviewer for more than 30 international famous academic journals.



Yi Wang (Member, IEEE) received the B.Eng. degree in electronic information engineering and the M.Eng. degree in information and signal processing from the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China, in 2013 and 2016, respectively, and the Ph.D. degree from the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore, in 2021.

He is currently a Research Assistant Professor with the Department of Electrical and Electronic Engineering, Hong Kong Polytechnic University, Hong Kong. His research interests include image restoration, image recognition, object detection and tracking, and crowd analysis.



Jiawei Lian (Graduate Student Member, IEEE) received the B.Eng. degree in automation from Jiangxi University of Science and Technology, Ganzhou, China, in 2019, and the M.Eng. degree in control engineering from Northwestern Polytechnical University, Xi'an, China, in 2022, where he is currently pursuing the Ph.D. degree in information and communication engineering with the School of Electronics and Information.

His research interests include trustworthy machine learning and remote sensing.



Zonghao Han received the B.S. degree in communication engineering from Ningxia University, Yinchuan, China, in 2020. He is currently pursuing the Ph.D. degree in information and communication engineering with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China.

His major research interests include hyperspectral remote sensing image processing, image-to-image translation, and deep learning.



Xiaoning Chen received the B.S. degree in electronic information science and technology and the M.S. degree in communication and information systems from Northwest University, Xi'an, China, in 2004 and 2011, respectively. She is currently pursuing the Ph.D. degree in information and communication engineering with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an.

Her research interests include computer vision and remote sensing image analysis.

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